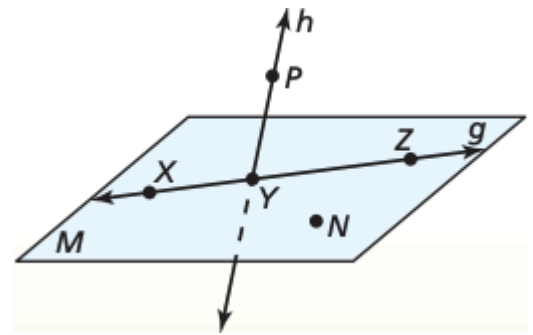


Chapter 1 Review

1.1 Understanding Points, Lines, and Planes

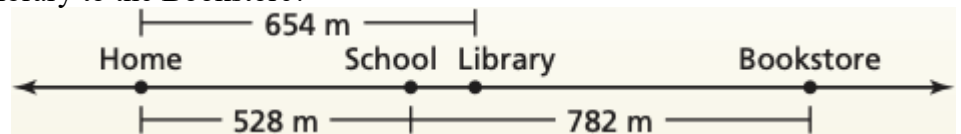
Use the diagram.

1. Give another name for Plane M .
2. Name a line in Plane M .
3. Give another name for that same line.
4. Name two rays.



1.2 Measuring Segments

5. How many meters is it from the Library to the Bookstore?



6. Let $FG = 4x - 9$, $GH = 3x + 1$, and $FH = 34$. Find FG and GH .
 - a. Set up an equation that represents the relationship in the diagram.
 - b. Solve for x .
 - c. Find FG and GH .

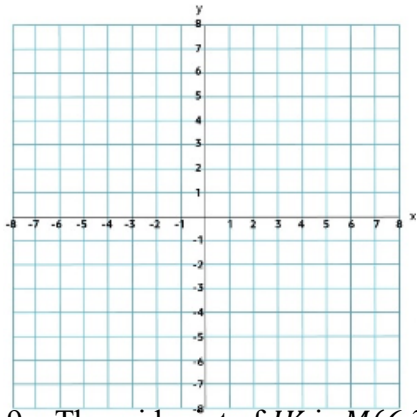


1.3 Using Midpoint and Distance Formulas

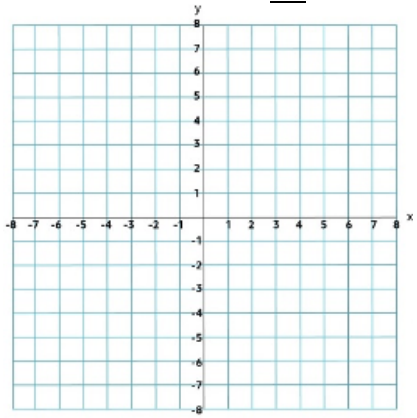
7. Let $AM = 3x + 8$ and $MB = 6x - 4$. Find AB .
 - a. Set up an equation that represents the relationship in the diagram.
 - b. Solve for x .
 - c. Find AB .



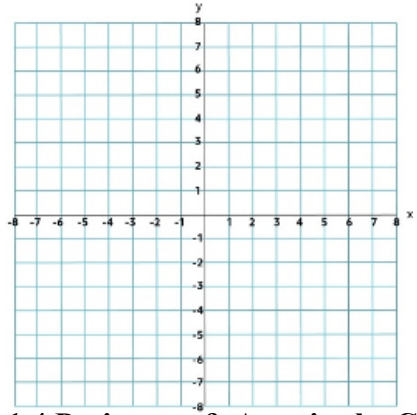
8. The endpoints of $\underline{ST}$ are $S(6, -3)$ and $T(7, -2)$. Find the coordinates for midpoint M .



9. The midpoint of $\underline{JK}$ is $M(6,3)$. One endpoint is $J(14,9)$. Find the coordinates of endpoint K .

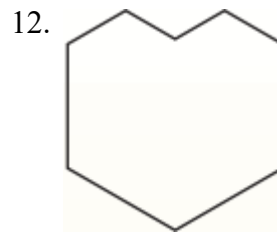
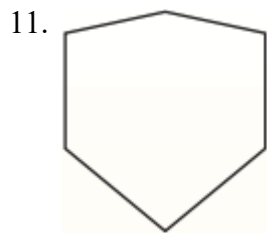


10. Find the distance between the two points $D(-2,4)$ and $E(3,9)$.



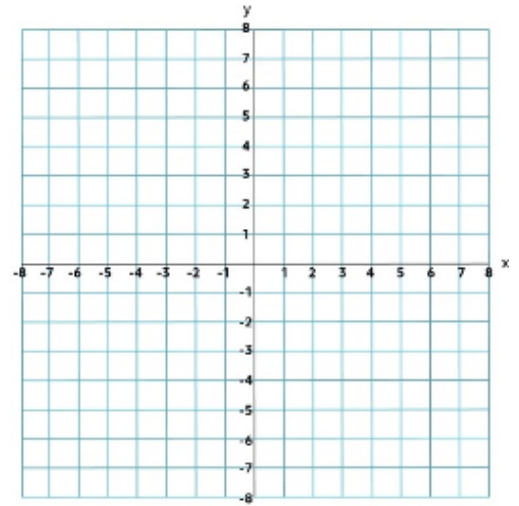
1.4 Perimeter & Area in the Coordinate Plane

Classify the polygon by the number of sides.



13. Find the perimeter and area from the polygon with vertices $W(-1, -5)$, $X(-5, -2)$, $Y(1, 6)$, and $Z(5, 3)$.

a. Find the segment lengths:



$WX = \underline{\hspace{2cm}}$ $XY = \underline{\hspace{2cm}}$ $YZ = \underline{\hspace{2cm}}$ $ZW = \underline{\hspace{2cm}}$

b. Find the Perimeter of $WXYZ$.

c. Find the Area of $WXYZ$.

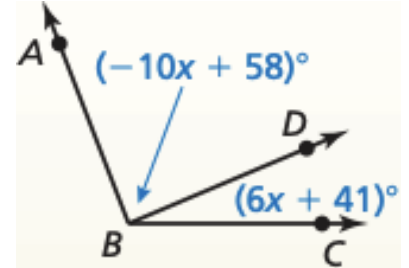
1.5 Measuring Angles

14. Let $m\angle ABC = 111^\circ$. Find $m\angle ABD$ and $m\angle DBC$.

a. Set up an equation that represents the relationship in the diagram.

b. Solve for x .

c. Find $m\angle ABD$ and $m\angle DBC$.

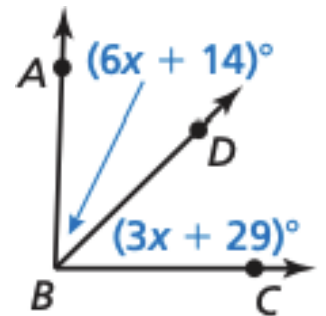


15. Let $\overrightarrow{BD}$ bisect $\angle ABC$. Find $m\angle ABD$ and $m\angle ABC$.

a. Set up an equation that represents the relationship in the diagram.

b. Solve for x .

c. Find $m\angle ABD$ and $m\angle ABC$.



1.6 Describing Pairs of Angles

Find the measure of each angle.

16. $\angle UVW$ and $\angle XYZ$ are complementary angles, $m\angle UVW = (x - 10)^\circ$, and $m\angle XYZ = (4x - 10)^\circ$.

a. Set up an equation that represents the relationship in the situation.

b. Solve for x .

c. Find $m\angle UVW$ and $m\angle XYZ$.

17. $\angle EFG$ and $\angle LMN$ are supplementary angles, $\angle EFG = (3x + 17)^\circ$, and $m\angle LMN = (\frac{1}{2}x - 5)^\circ$.

a. Set up an equation that represents the relationship in the situation.

b. Solve for x .

c. Find $m\angle EFG$ and $m\angle LMN$.

18. Match the angle pair with its description.

a. Linear Pair

b. Vertical Angles

c. Nonadjacent Angles

$\angle 2$ & $\angle 4$: _____.

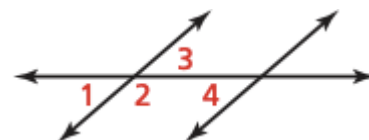
$\angle 1$ & $\angle 3$: _____

$\angle 2$ & $\angle 3$: _____

$\angle 4$ & $\angle 1$: _____.

$\angle 1$ & $\angle 2$: _____

$\angle 3$ & $\angle 4$: _____



19. Two angles are complementary. The measure of the larger angle is 14° more than 3 times the measure of the smaller angle. Find the measure of both angles.

a. Set up an equation that represents the relationship described.

b. Find the smaller angle's measure.

c. Find the larger angle's measure.